Weekly Assignment: Arguments & Circuits (Epp 2.3 & 2.4)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 2.3 #9). Use a truth table to determine whether the following argument form is valid. Indicate which columns represent the premises and which represent the conclusion, and include a sentence explaining how the truth table supports your answer.

$$\begin{array}{l} p \vee q \to \neg r \\ p \vee \neg q \\ \neg q \to p \\ \hline \therefore\ \neg r \end{array}$$

Problem 2 (Epp 2.3 #28). Determine whether the following argument is valid or invalid. Use symbols to write the logical form of the argument. If the argument is valid, identify the rule of inference that guarantees its validity. Otherwise, state whether the converse or inverse error is made.

> If there are as many rational numbers as there are irrational numbers, then the set of all irrational numbers is infinite.
> The set of all irrational numbers is infinite.
> ∴ There are as many rational numbers as there are irrational numbers.

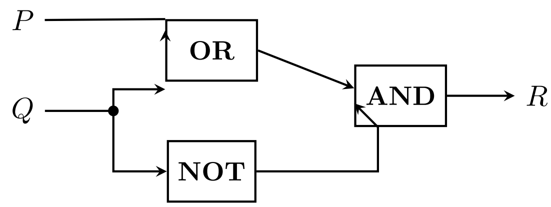
Problem 3 (Epp 2.3 #30). Determine whether the following argument is valid or invalid. Use symbols to write the logical form of the argument. If the argument is valid, identify the rule of inference that guarantees its validity. Otherwise, state whether the converse or inverse error is made.

> If this computer program is correct, then it produces the correct output when run with the test data my teacher gave me.
> This computer program produces the correct output when run with the test data my teacher gave me.
> ∴ This computer program is correct.

Problem 4 (Epp 2.3 #44). Use the valid argument forms listed in Table 2.3.1 to deduce the conclusion from the premises, giving a reason for each step as in Example 2.3.8. Assume all variables are statement variables.

$$\begin{array}{rl} \text{a.} & p \to q \\ \text{b.} & r \vee s \\ \text{c.} & \neg s \to \neg t \\ \text{d.} & \neg q \vee s \\ \text{e.} & \neg s \\ \text{f.} & \neg p \wedge r \to u \\ \text{g.} & w \vee t \\ \hline & \therefore\ u \wedge w \end{array}$$

Problem 5 (Epp 2.4 #2). Give the output signal for the following circuit if the input signals are $P = 1$ and $Q = 0$.



The circuit has inputs P and Q. The signal Q branches: one copy goes (together with P) into an **OR** gate, the other into a **NOT** gate. The outputs of the OR gate and the NOT gate then feed into an **AND** gate, producing the output R.

Problem 6 (Epp 2.4 #19). For the input/output table below, construct

(a) a Boolean expression having the given table as its truth table, and

(b) a circuit having the given table as its input/output table.

P	Q	R	S
1	1	1	0
1	1	0	1
1	0	1	0
1	0	0	1
0	1	1	0
0	1	0	1
0	0	1	0
0	0	0	0